Incorporating Geometric Priors in Semantic Mapping

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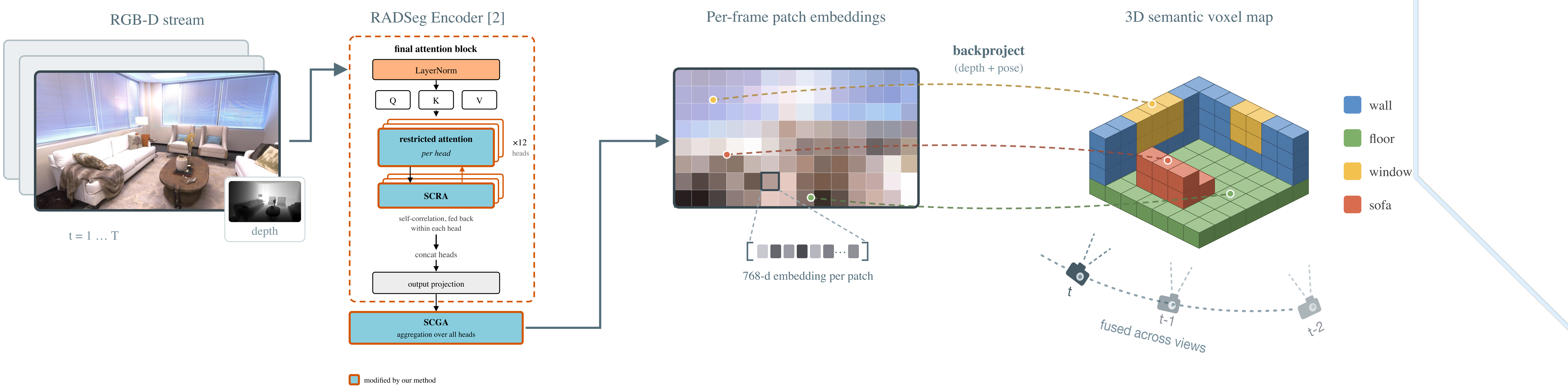


Setting. Semantic voxel mapping labels a scene by encoding each posed RGB-D frame in 2D and fusing the per-patch features into a 3D voxel grid, so every error in the 2D encoder is baked into the map.

Problem. The quality of segmentation declines as attention is paid to the final mix of patches across class boundaries [1]. Using SAM [3] masks to limit attention is a remedy, but is slow to run on every frame of a mapping sequence.

Ours. Depth is already available in the mapping. We restrict attention to planar surfaces and 3D locality fitted directly from the depth map, at a small cost.

Methods

Step 1: geometric partition. We backproject the depth map and fit planes with CAPE, giving surfaces $\mathcal{S} = \{S_1, \ldots, S_K\}$. Each patch i is assigned to a plane if enough of its support lies within τ of that plane, and left unassigned otherwise:

$$s(i) \in \{1, \ldots, K\} \cup \{\varnothing\}.$$

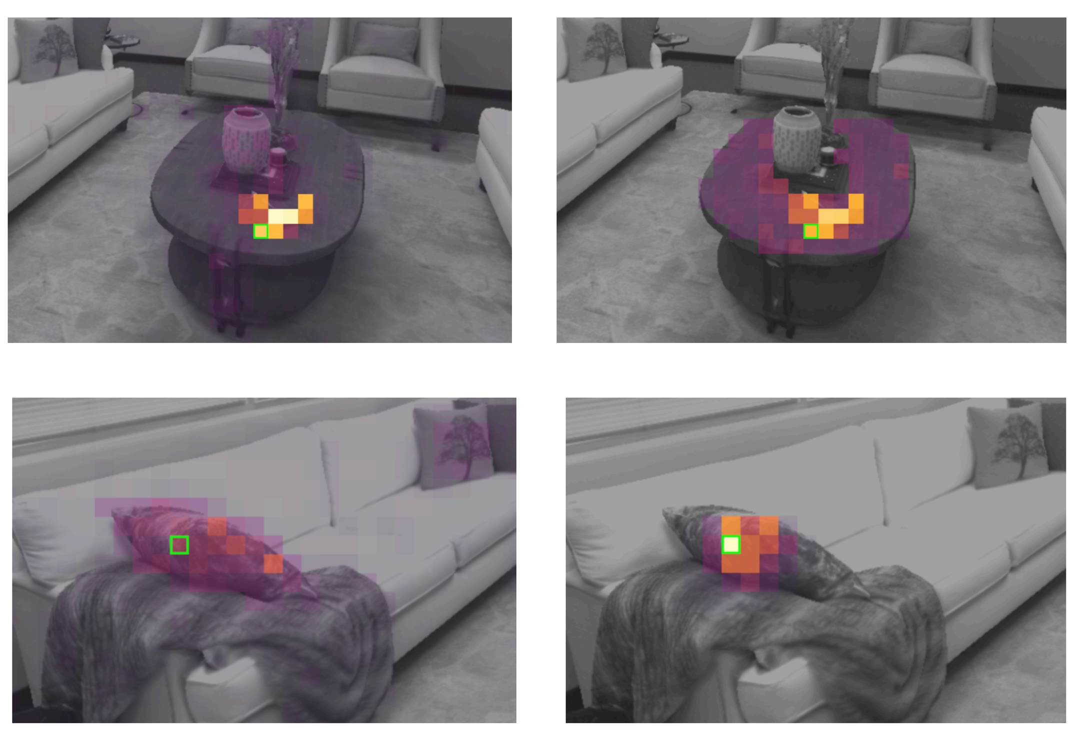
Patches straddling two planes or lying on unmodeled geometry receive $s(i) = \varnothing$.

Step 2: restricted correlation. In the final attention block a geometric bias is added to the logits,

$$Attention = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}} + B\right) V,$$

$$B_{ij} = \begin{cases} 0, & s(j) = s(i) \neq \varnothing, \\ -\infty, & s(i) \neq \varnothing,\ s(j) \neq s(i), \\ -\dfrac{\|p_i - p_j\|^2}{2\sigma^2} & s(i) = \varnothing. \end{cases}$$

A query on a surface sees only that surface. A query on no surface has no region to restrict to. It has visibility to one averaged K per each surface, and every unassigned patch's K. Those patches receive a soft surface 3D distance kernel prior, zero for a co-located, strictly negative otherwise. Here σ is the trust radius in metres. A similar design is used to limit feature mixing in RADSeg's [2] SCRA and SCGA.



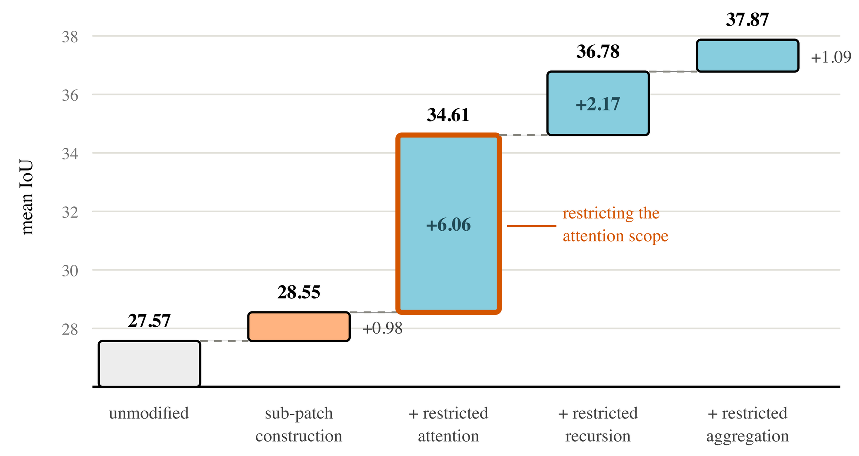
Top row: Regular vs. our attention for a surface (table) query patch.
Bottom row: Regular vs. our attention for an unassigned (pillow) query patch.
A surface gives a proxy for semantic membership; for unassigned patches, proximity is a weaker but vocabulary-free fallback.

Results

Method	Attn	SCRA	SCGA	Resolution	ScanNet mIoU	ScanNet fmIoU	ScanNet Time	Replica mIoU	Replica fmIoU	Replica Time
	method applied on				ScanNet			Replica		
RADSeg (baseline)	✗	✗	✗	native	40.1	48.8	**1.00×**	33.1	60.5	**1.00×**
baseline with subpatches	✗	✗	✗	upsampled	40.2	48.6	2.69×	34.3	61.5	2.61×
Ours (attn)	✓	✗	✗	native	**41.6**	<u>51.1</u>	<u>1.86×</u>	35.0	63.7	1.65×
Ours (attn)	✓	✗	✗	upsampled	<u>41.5</u>	**51.2**	3.02×	35.9	**65.3**	2.98×
Ours (SCRA)	✗	✓	✗	native	41.0	50.6	<u>1.86×</u>	35.4	63.2	<u>1.64×</u>
Ours (SCRA)	✗	✓	✗	upsampled	41.1	50.7	2.99×	<u>36.1</u>	64.5	2.93×
Ours (all)	✓	✓	✓	native	41.4	50.8	1.90×	34.5	63.4	1.70×
Ours (all)	✓	✓	✓	upsampled	41.3	<u>51.1</u>	3.31×	**36.2**	<u>65.2</u>	3.34×

When our method is applied to near-perfect depth with Replica, it performs best when injected at all 3 possible points (attn, SCRA, SCGA) and with upsampled features. With ScanNet's noisy depth, the best mode uses only the first injection point (attn) and native patch resolution. SCRA and SCGA likely "cleanup" and prevent compounded semantic locality mistakes in that case. PriorDepthAnything [4] is used to support ScanNet's depth map; the FPS estimates include this, and all other operations for end-to-end time per keyframe processed.

Where the geometric prior helps, and where it cannot. Left to right: RGB frame, our prediction, and a change map against the baseline. Green marks pixels we fix, red those we break, and blue those both models get wrong. Green dominates the object boundaries, where restricted attention stops features from bleeding across surfaces. The blinds are the failure case: they are coplanar with the window, so a single fitted plane spans both and the blinds are absorbed into the window class.



The improvement over baseline in 2D segmentation is due to attention mechanism, not resolution.
Components added cumulatively, left to right; each bar is that component's contribution to mean IoU. Splitting patches into depth-interpolated subpatches alone buys almost nothing — the gain arrives when the attention scope is restricted to the same regions. ADE20K validation, all 2000 images, ground-truth regions, identical readout throughout.

Next steps

Sharper locality for unassigned patches. Add normal agreement to the 3D kernel, so a crease between two touching surfaces separates patches that is not currently penalized.

Depth-aware fusion. Down-weight patches whose backprojected depth disagrees across views, so voxels are not corrupted by boundary and patches split over objects.

Appearance where geometry is blind. Coplanar objects are indistinguishable to depth but often separable in feature space, so clustering patches within each fitted plane could recover object boundaries that the plane itself cannot.

[1] D. Zhang, F. Liu, Q. Wang. CorrCLIP: Reconstructing Patch Correlations in CLIP for Open-Vocabulary Semantic Segmentation. *ICCV*, 2025. [2] O. Alama, D. Jariwala, A. Bhattacharya, S. Kim, W. Wang, S. Scherer. RADSeg: Unleashing Parameter and Compute Efficient Zero-Shot Open-Vocabulary Segmentation Using Agglomerative Models. *CVPR Findings*, 2026.
[3] A. Kirillov et al. Segment Anything. *ICCV*, 2023. [4] Z. Wang, S. Chen, L. Yang, J. Wang, Z. Zhang, H. Zhao, Z. Zhao. Depth Anything with Any Prior. *arXiv:2505.10565*, 2025.